O-B-420a
December 2, 1964
SUPERSEDING
Fed. Spec. O-B-420
April 23, 1962
(See 6.4)

FEDERAL SPECIFICATION

BLEACH, LAUNDRY, ORGANIC CHLORINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type and one grade of organic chlorine, dry laundry bleach (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378—Plastic Film (Polyethylene Thin Gage).
P-D-245—Detergent (Synthetic) Built, High and Low Sudsing.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-D-723—Drums, Fiber.
PPP-T-60—Tape; Pressure Sensitive Adhesive, Waterproof—For Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date

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of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) *Designations*:

E 70-52T—Method for Determination of pH of Aqueous Solutions With the Glass Electrode (Tentative).

D 2022-62T—Methods of Sampling and Chemical Analysis of Chlorine Containing Bleaches (Tentative).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Federal Hazardous Substances Labeling Act (Public Law 86-613, 86th Congress 5.1283 approved July 12, 1960).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material. The laundry bleach shall be a white product in powder or bead form containing an organic, chlorine-liberating compound.

3.2 Chemical and physical requirements. The product shall conform to the requirements specified in table I when tested as specified in 4.3.2.

TABLE I—*Requirements*

	Minimum	Maximum
Available chlorine, percent	15.0	17.0
Phosphates, as P_2O_5, percent[1]	14.0	—
Silicates as SiO_2, percent	1.0	5.0
Carbonates, as Na_2CO_3, percent	—	0.5
Reaction with carbonate ion, turbidity or precipitation	None	
pH of an aqueous solution	8.0	11.0
Insoluble matter content, percent[2]	—	1.5
Moisture content, percent	—	2.0
Loss on ignition, percent	18.0	—
Thermal decomposition point, degrees centigrade	200	—
Loss of available chlorine on heating, percent	—	20.0
Bulk density, grams per milliliter	—	1.1
Particle size, pass through:		
16 mesh sieve, percent	90	—
100 mesh sieve, percent	—	20
200 mesh sieve, percent	—	5

[1] The phosphate content shall be made up of sodium tripolyphosphate or tetrasodium pyrophosphate or a mixture of these. The supplier shall submit a certificate of compliance for this characteristic. The certificate shall be accompanied by actual test, inspection or other verifiable quality data.

[2] To include dedusting agent, if present.

3.3 Performance. The bleach shall remove not less than 78 percent of the stain when tested as specified in 4.3.2.

3.4 Workmanship. The material shall be clean, uniform, free flowing, shall show no evidence of segregation of individual constituents or of lumping and caking and no foreign material shall be present. The occurrence of defects shall not exceed the applicable acceptable quality levels in section 4.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Sampling for inspection.* Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all material from the same manufactured batch.

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4.2 **Inspection of materials and components.** The supplier is responsible for insuring that all materials and components used were inspected in accordance with the requirements of this specification and of referenced specifications.

4.3 **Inspection of the end item.**

4.3.1 *Examination of the end item.* The end item shall be examined for the defects in applicable sub-paragraphs and at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.5. A lot shall consist of laundry bleach manufactured in one batch. The lot shall be expressed in units of filled unit containers of the same capacity for examinations in 4.3.1.1, 4.3.1.2, 4.3.1.3, and in units of shipping containers for examination for preparation for delivery under 4.3.1.4.

4.3.1.1 *Examination of the unit container.* The sample unit for this examination shall be one filled unit container.

Examine	Defect
Construction	Not as specified (see 5.1).
Workmanship	Dented, crushed, cracked, abraded, dirty, or torn.
Closure	Incomplete, imperfect, leakage or damage to polyethylene liner.
Markings	Missing, incomplete, incorrect. Not as specified (see 5.3).
Liner	Missing. Not as specified (see 5.1.1); punctured or damaged. Not completely heat sealed at seams, knotted, or not double folded and taped (as applicable).
Packing	Not in accordance with requirements (see 5.2).

4.3.1.2 *Examination of contents of the unit container.* The sample unit for this examination shall be the contents of one filled unit container.

Examine	Defect
Appearance	Presence of foreign material. Not uniform. Not free flowing. Contains agglomerates, cakes, or lumps. Not clean.

4.3.1.3 *Examination of the end item for net contents.* The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container is less than specified or indicated for all sample units examined.

4.3.1.4 *Examination of preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, and contents comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. This examination does not apply when the unit container is the shipping container.

Examine	Defect
Markings	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components: such as incomplete closure of case liners, container flaps; loose or inadequate sealing, strapping, or stapling. Bulged or distorted containers.
Contents	Number of intermediate packages (where applicable) per shipping container is more or less than required. Intermediate container (where applicable) not as specified.

4.3.1.5 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels, for determining the sample size, and the acceptable quality lev-

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els (AQLs) expressed in defects per one hundred units shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.1.1	S-3	2.5
4.3.1.2	S-3	2.5
4.3.1.3	S-3	—
4.3.1.4	S-2	4.0

4.3.2 *Testing of the end item.* The methods of testing specified in Fed. Test Method Std. No. 536, wherever applicable and as listed in table II, shall be followed. The sample unit for testing shall be as specified in 4.3.2.1 and 4.3.2.2. The lot size shall be expressed in pounds of bleach. The sample size shall be as shown below. When specified in table II, tests shall be performed on a composite sample or sample unit basis. The lot shall be unacceptable if one or more sample units or the composite sample fail to meet any test requirement specified.

Lot size (pounds)	Sample size
300 or less	2
301 to 22,000	3
22,001 or more	5

4.3.2.1 The sample for composite testing shall be two pounds of bleach, selected in equal portions from each sample unit. The portions from the individual samples shall be thoroughly mixed, yielding the composite. The sample shall be placed in an airtight and watertight brown glass container with suitable caps, covered, and sealed, to prevent any moisture change.

4.3.2.2 The sample unit for testing on a sample unit basis shall be 5 ounces, drawn in segregated 1-ounce portions and placed into 5 individual 1-ounce brown glass airtight and watertight jars, covered, and sealed to prevent any moisture change. A separate 1-ounce portion shall be used for each test made on a sample unit basis.

TABLE II—*Instructions for testing of the end item*

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as	
	Requirement	Test method	Individ. unit	Composite sample		Pass or fail	Numerically to nearest
Available chlorine, percent	Table I	4.4.1	X		Average of 2		0.1
P_2O_5 content, percent	Table I	1501		X	Average of 2		0.1
SiO_2 content, percent	Table I	1301	X		Average of 2		0.1
CO_2 as Na_2CO_3 content, percent	Table I	4.4.2		X	Average of 2		0.1
Reaction with carbonate ion, turbidity, or precipitation	Table I	4.4.3		X	Average of 2	X	
pH of an aqueous solution	Table I	4.4.4		X	Average of 2		0.1
Insoluble matter content, percent	Table I	4.4.5	X		Average of 2		0.1
Moisture content, percent	Table I	4.4.6	X		Average of 2		0.1
Loss of ignition	Table I	4.4.7		X	Average of 3		0.1
Thermal decomposition point, degrees centigrade	Table I	4.4.8		X	Average of 2		1
Loss of available chlorine on heating, percent	Table I	4.4.9	X		Average of 2		0.1
Bulk density, grams per milliliter	Table I	4.4.10		X	1		0.1
Particle size: Pass through							
16 mesh sieve, percent	Table I	2101		X	1		1.0
100 mesh sieve, percent	Table I	2101		X	1		1.0
200 mesh sieve, percent	Table I	2101		X	1		1.0
Performance, percent	3.3	4.4.11		X	1		1.0

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4.4 **Test procedures.** Bleach sample stored in brown glass containers shall be tightly sealed and shall not be exposed to sunlight or to the atmosphere during the course of analyses. Prior to each analysis, except where otherwise specified, a sample of bleach, approximately double the weight required shall be thoroughly ground and mixed in a dry mortar. The sample required for the analysis shall be immediately weighed from the ground sample.

4.4.1 *Available chlorine.*

4.4.1.1 *Reagents.*

a. Sodium thiosulfate ($Na_2S_2O_3$), ACS grade.
b. Hydrochloric acid (HCl), ACS grade.
c. Potassium iodide (KI), ACS grade.
d. Sodium hydroxide (NaOH), ACS grade.
e. Soluble starch.

4.4.1.2 *Solutions.*

a. Sodium thiosulfate 1.00N.
b. Hydrochloric acid 1:1.
c. Alkaline potassium iodide (200 grams KI and 120 grams NaOH per liter).
d. Starch solution, 1 percent.

4.4.1.3 *Procedure.* Weigh 10.000 ± 0.001 g. of ground and mixed bleach sample into a dry 500-ml. Erlenmeyer flask containing a magnetic stirring bar. Add 200 ml. of distilled water and 50 ml. of alkaline potassium iodide solution. Agitate for at least 5 minutes to effect solution of the bleach, then slowly add 50 ml. of hydrochloric acid solution while continuing to agitate. (Rapid addition of the acid will cause too rapid release of the free iodine and presence of iodine vapors above the solution in the flask. Loss of iodine vapors to the atmosphere will give erroneous results.) Titrate immediately with thiosulfate solution to a straw color. Add 5 ml. of starch solution and continue titration slowly to the disappearance of blue color.

4.4.1.4 *Calculation.*

$$\text{Percent available chlorine} = \frac{\text{ml. } Na_2S_2O_3 \times \text{N of } Na_2S_2O_3 \times 0.03546 \times 100}{\text{Weight of sample (g)}}$$

4.4.2 *Carbonates as Na_2CO_3.* Carbonate content expressed as Na_2CO_3 shall be determined in accordance with method 1401 of Fed. Test Method Std. No. 536 except that the fifth sentence under "Procedure" shall be modified to read as follows: "Weigh to the nearest 0.001 g. a sufficient amount of ground and mixed bleach sample to yield approximately 0.2 g. of CO_2, transfer the bleach to the Knorr flask, cover with 20 ml. of freshly boiled distilled water, add hydroxylamine sulfate (approximately equal in weight to the bleach sample—use flask large enough to accommodate sample and reagents and increase size of tube carrying acid into the sample flask as required) and close the apparatus with the entire train in place". Continue with sentence number six under "Procedure".

4.4.3 *Reaction with carbonate ion.* Dissolve 1.00 ± 0.01 g. of ground and mixed bleach sample in 100 ml. of distilled water (at 25° ± 2°C). Filter the solution to remove insoluble material and to obtain a clear solution. Add to this solution 2.00 ± 0.02 g. of anhydrous sodium carbonate (Na_2CO_3). Stir or shake this system for five minutes. Allow to stand for an additional five minutes. The solution shall be as clear as a control solution consisting of 2.00 ± 0.02 g. of anhydrous sodium carbonate in 100 ml. of distilled water when visually observed in matched Nessler tubes.

4.4.4 *Determination of pH.* Determine the pH at 25° ± 1°C. of a solution containing 200 parts per million available chlorine as prepared in 4.4.11.2 using a pH meter with an alkali-resistant glass electrode in accordance with ASTM Designation E 70-52T.

4.4.5 *Determination of insoluble matter content.* Weigh out 0.312 ± .001 g. of ground and mixed bleach sample. Place 500 ml. of

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distilled water in a 1000-ml. beaker and raise the temperature of the water to 65° ± 2°C. While stirring the water with a mechanical stirrer, add the weighed bleach into the beaker. Continue to stir slowly for 5 minutes. Filter the solution and precipitate, if any, into a tared Gooch or fritted glass crucible. Wash the insolubles with three 20-ml. portions of distilled water. Dry the insolubles in an oven at 105°C. to constant weight.

4.4.5.1 *Calculation.* Calculate percent weight of the insolubles as follows:

$$\text{Weight percent insolubles} = \frac{A - B}{C} \times 100$$

A = Weight of crucible plus insolubles.
B = Weight of crucible.
C = Weight of sample of laundry bleach.

4.4.6 *Moisture content.* The moisture content shall be determined in accordance with ASTM Designation D 2022-62T, sections 34 through 39. The sample designated in 4.3.2.2 ground and mixed shall be used for the moisture determination.

4.4.7 *Loss on ignition.* Place one gram (1.000 ± .001) of ground and mixed bleach sample in a tared porcelain crucible. The crucible with its contents is set vertically in a triangle on a tripod and heated over a Meker-type burner for one hour. The crucible with its residual contents (after ignition) is cooled under desiccation and weighed.

4.4.7.1 *Calculation.*

$$\text{Percent loss on ignition} = \frac{(C + W - D)}{W} \times 100$$

C = Weight of crucible.
W = Weight of bleach sample.
D = Weight of residue and crucible after ignition.

4.4.8 *Thermal decomposition point.* A small Wood's metal bath shall be prepared with a 250-ml. metal beaker filled to approximately three-fourths capacity with molten Wood's metal. This bath shall be supported above a Bunsen burner and a thermometer (0° to 360°C.) shall be inserted in the Wood's metal one inch from the side of the beaker so that the top of the mercury bulb is one inch below the surface. A safety shield shall then be placed in front of this assembly. The analyst should wear protective clothing and complete eye and face protection. Into a 1- by 8-inch Pyrex test tube 5.0 grams of the unground product shall be placed. The bulb of a thermometer, 0° to 360°C., shall be immersed in this sample and the test tube shall be supported over the Wood's metal bath so that the top of the sample is just below the level of the Wood's metal. Heat shall be applied to the metal bath at such a rate that the temperature of the sample under test does not lag more than 10°C. below the temperature of the bath. The temperature of the bleach shall be watched constantly. Visual indications of rapid reaction such as mild explosion, fire or sudden rapid rise to a temperature greater than that of the bath shall be considered as evidence of thermal decomposition. The temperature taken as the point of decomposition is not the temperature reached after reaction has taken place but is rather the last observed temperature before the reaction.

4.4.9 *Loss of available chlorine on heating.* Place 5.0000 ± 0.0005 g. of unground bleach sample into each of two test tubes (150 mm. long and 25 mm. in diameter), and insert rubber stoppers carrying air condensers (glass tubes) 15 inches long and 1/4 inch in internal diameter. The tubes shall extend 1/2 inch beyond the stopper into the test tubes. Place the test tubes in a constant temperature bath at 80° ± 2°C. for 6 hours. At the end of the heating period, remove the tubes, substitute solid rubber stoppers for the air condenser stoppers, and allow to cool to room temperature. Determine the available chlorine by the method specified in 4.4.1 combining the two samples to perform this determination. In calculating the percent of available chlorine after heating, use the weight of the combined samples taken before heating.

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4.4.9.1 *Calculation.*

$$\text{Percent loss of available chlorine after heating} = \frac{(A - B) \times 100}{A}$$

Where: A = Percent available chlorine as received.
B = Percent available chlorine after heating.

4.4.10 *Bulk density.* Weigh 40.00 ± 0.01 g. of unground sample into a 100-ml. graduated cylinder from which the lip has been removed. Stopper the graduate and pass a closely fitting glass sleeve, about 3 inches long, over it. Clamp the sleeve to a ringstand. Place a large rubber stopper under the cylinder and adjust the sleeve so that the graduate will be 10 centimeters above the rubber stopper when the base of the graduate touches the lower edge of the sleeve. Raise the graduate until it touches the sleeve, then release. Continue raising and dropping the graduate until 20 cycles are completed. Read the volume of the sample to the nearest milliliter and calculate the bulk density in grams per milliliter.

4.4.11 *Performance test.*

4.4.11.1 *Preparation of stained swatches.*

a. Cloth. The cloth shall be preshrunk, plain weave, permanent finish cotton and shall contain approximately 54 ends and 45 picks per square inch and weigh approximately 5.1 ounces per square yard. The initial reflectance of the cloth shall be determined using a Model 610 Photovolt or equivalent, and a green tri-stimulus filter. Thirty-two, 3.5- by 3.5-inch, swatches shall be prepared.

b. Staining. Bring one liter of tap water to a boil. Insert 5 tea bags (Orange Pekoe Cut Black) in the boiling water and boil for 5 minutes, then remove bags. Insert 32 cloth swatches (3.5 by 3.5 inches) and continue to boil for 5 minutes. Remove swatches, wring out and dry in a clothes dryer at 180° ± 10°F. for 30 minutes. Rinse in cold water and repeat drying procedure.

Note: Stained swatches older than 2 weeks shall be discarded and fresh swatches prepared. Record the reflectance of the stained swatches taking average readings of both sides. Use swatches stacked four deep for the readings in order to get a background of similar reflectance.

4.4.11.2 *Stain removal.*

a. Apparatus. A Launder-Ometer with pre-heating load table, or similar machine, shall be used, in which 1-pint glass jars are held with their bases toward a horizontal shaft and 2 inches from its center of rotation, the shaft having a speed of totation of 40 to 45 r.p.m. The temperature of the bath shall be maintained at 49° ± 2°C.

b. Reagents.

Water. The hardness of the water shall not exceed 50 p.p.m. calculated as calcium carbonate ($CaCO_3$).

Detergent solution. A 0.4 percent solution of a synthetic detergent (bead form) conforming to type I of P-D-245 shall be used.

Bleach solution. A solution of laundry bleach shall be prepared in a 2-liter volumetric flask using sufficient ground and mixed dry bleach to give a concentration of 200 p.p.m. available chlorine (0.02 percent). Weigh bleach sample directly into the flask from a tared container of bleach and determine sample weight by difference. Prepare this solution immediately before use.

c. Procedure. Measure into each of five 1-pint glass jars, each containing ten 1/4-inch stainless steel balls, 150 ml. of detergent solution and 150 ml. of bleach solution both solutions at a temperature of 49° C. (120°F.). Place two stained swatches in each jar. Cap the jars and place in washing machine. The machine is operated for 20 minutes at temperature of 49° ± 2°C. After washing, the swatches are removed, rinsed thoroughly in water at a temperature of 45°—50°C., squeezed by hand to remove excess water, smoothed flat, and air dried. Measure the reflectance of the bleach swatches, taking average readings on both sides. Use swatches stacked four deep for

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readings in order to get a background of similar reflectance.

4.4.11.3 *Calculation.* Calculate the total percent stain removed as follows:

$$\text{Total percent stain removed} = \frac{B - S}{O - S} \times 100$$

B = Average swatch reflectance after bleaching.
S = Average swatch reflectance of stained swatch.
O = Initial reflectance of original unstained swatch.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified, (see 6.2).

5.1.1 *Levels A and B.* Two pounds of the laundry bleach shall be packaged in a sift-proof folding paperboard box conforming to style I, type A, class a or b of PPP-B-566 or a commercially equivalent folding box. Each box shall be lined with a 0.002-inch thick polyethylene bag conforming to type I, grade A, finish 1 of L-P-378. All seams shall be accomplished by heat sealing and shall conform to the heat-seal-strength requirement in L-P-378. The bag closure shall be accomplished by heat sealing or double folding of the plastic material at the mouth and taping it to the bag body with a minimum one-inch width tape conforming to type III, class 2 of PPP-T-60. Prior to or during the final closure operation, excess air within the bag shall be expelled.

5.1.2 *Level C.* Two pounds of the laundry bleach shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C as specified, (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Two-pound boxes.* Twenty-four 2-pound boxes of laundry bleach, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to V3c or V3s, style RSC or PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.1.2 *Bulk quantity.* The laundry bleach shall be bulk packed in 25-, 50, or 125-pound quantities as specified (see 6.2).

5.2.1.2.1 *Twenty-five pound quantity.* Twenty-five pounds of laundry bleach shall be packed in a fiberboard shipping container conforming to V3c or V3s, style FOL of PPP-B-636. Each shipping container shall be lined with a 0.004-inch thick polyethylene bag conforming to type 1, grade A, finish 1 of L-P-378. All seams shall be accomplished by heat sealing and shall conform to the heat-seal-strength requirements in L-P-378. The bag body shall be of sufficient length so that after removal of portions of the contents, the remaining contents may be closed off by tieing a loose knot with the bag. The bag closure shall be accomplished by heat sealing, tightly knotting the plastic material at the mouth, or double folding of the plastic material at the mouth, and taping it to the bag body with a minimum one-inch width tape conforming to type III, class 2 of PPP-T-60. Prior to or during the final closure operation, excess air within the bag shall be expelled. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. When the bottom flaps of the shipping container are seamed by metal stitching, the container shall be provided with a bottom pad fabricated from chipboard or fiberboard. The length and width of the pad shall be approximately one-eighth inch less than the inside length and width of the container.

5.2.1.2.2 *Fifty or one-hundred and twenty-five-pound quantities.* Fifty or 125 pounds

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of laundry bleach shall be packed in a fiber drum conforming to type III, grade A of PPP-D-723. Each drum shall be lined with a polyethylene bag and the bag shall be sealed and closed as specified in 5.2.1.2.1.

5.2.2 *Level B.*

5.2.2.1 *Two-pound boxes.* Twenty-four two-pound boxes of laundry bleach, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, variety optional of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.2.2.2 *Bulk quantity.* The laundry bleach shall be bulk packed in 25-, 50-, or 125-pound quantities as specified (see 6.2).

5.2.2.2.1 *Twenty-five-pound quantity.* Twenty-five pounds of laundry bleach shall be packed in a fiberboard shipping container conforming to style FOL, type CF or SF, class domestic, variety optional of PPP-B-636. Each shipping container shall be lined with a polyethylene bag and the bag shall be sealed and closed as specified in 5.2.1.2.1. Closure shall be in accordance with the appendix of the container specification. When the bottom flaps of the shipping container are seamed by metal stitching, the container shall be provided with a bottom pad fabricated from chipboard or fiberboard. The length and width of the pad shall be approximately one-eighth inch less than the inside length and width of the container.

5.2.2.2.2 *Fifty or one-hundred and twenty-five-pound quantities.* Fifty or 125 pounds of laundry bleach shall be packed in a fiber drum conforming to type I, grade A, class 1 of PPP-D-723. Each drum shall be lined with a polyethylene bag and the bag shall be sealed and closed as specified in 5.2.1.2.1.

5.2.3 *Level C.* Laundry bleach, packaged as specified in 5.1, or 25-, 50-, or 125-pound quantities of laundry bleach as specified (see 6.2) shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Labeling.* Unless otherwise specified in the contract or order, each container shall be labeled in accordance with MIL-STD-129, and the Federal Hazardous Substances Labeling Act. The following legend shall also be included:

DIRECTIONS FOR USE

Dry bleach may be used in white cotton wash formulas with either soap or synthetic detergents. During bleach operation add dry bleach to wash-wheel in the following quantities:

Average soil—2 ounces per 100 lb. load.
Medium soil—3 ounces per 100 lb. load.
Heavy soil—4 ounces per 100 lb. load.
Keep container tightly closed.
Keep out of contact with combustible materials and laundry sours.
Store in cool, dry place.
Date of packaging.

5.3.1.1 *Warning legend.* Directly beneath "DIRECTIONS FOR USE" add, in red ink, the following legend:

WARNING

THIS PRODUCT IS NOT TO BE USED FOR THE DISINFECTION OF WATER OR FRUITS OR VEGETABLES.

5.3.2 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 *Military requirements.* In addition to any special marking required by the con-

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tract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dry laundry bleach covered by this specification is intended for direct addition to the laundry wheel in laundering operations. This product is not intended to be used for the disinfection of water or fruits or vegetables.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(*c*) Weight of bulk pack required (see 5.2.1.2, 5.2.2.2 and 5.2.3).

6.3 The laundry bleach must be stored in a cool, dry place. Stocks should be rotated and overstocking avoided. Not more than six month's supply should be purchased at one time. When not in use, the drums should be kept tightly closed.

6.4 Supersession data. This specification includes the requirements of Military Specification MIL-B-22743 (Ships) dated January 18, 1961.

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are as follows:

Rail:

Bleach, Laundry, dry, not otherwise indexed by name.

Carload minimum weight 36,000 pounds.

Motor:

Bleach, Laundry, dry, not otherwise indexed.

Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—SH

Air Force—68

Review activities:

Army—GL, MD, MU

Navy—SH, MS

Air Force—68

User activities:

Army—SM

Navy—MC

Preparing activity:

Army—GL

Reviewer and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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